

TROWBRIDGE, ON, Canada

WMO: 714611

Lat: 48.292N

Lon: 88.876W

Elev: 218

StdP: 98.73

Time Zone: -5.00 (NAE)

Period: 88-00

WBAN: 99999

Annual Heating, Humidification, and Ventilation Design Conditions

Coldest Month	Heating DB		Humidification DP/MCDB and HR						Coldest Month WS/MCDB				MCWS/PCWD to 99.6% DB		WSF
			99.6%			99%			0.4%		1%				
	99.6%	99%	DP	HR	MCDB	DP	HR	MCDB	WS	MCDB	WS	MCDB	MCWS	PCWD	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
1	-24.6	-22.1	-30.7	0.2	-21.1	-28.5	0.3	-20.2	20.3	-2.3	19.0	-6.4	8.6	310	0.877

Annual Cooling, Dehumidification, and Enthalpy Design Conditions

Hottest Month	Hottest Month DB Range	Cooling DB/MCWB						Evaporation WB/MCDB						MCWS/PCWD to 0.4% DB	
		0.4%		1%		2%		0.4%		1%		2%			
		DB	MCWB	DB	MCWB	DB	MCWB	WB	MCDB	WB	MCDB	WB	MCDB	MCWS	PCWD
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
8	4.8	22.2	15.1	20.6	15.0	19.3	14.8	17.6	19.9	16.7	19.1	15.9	18.3	6.4	320

Dehumidification DP/MCDB and HR									Enthalpy/MCDB						Extreme Max WB
0.4%			1%			2%			0.4%		1%		2%		
DP	HR	MCDB	DP	HR	MCDB	DP	HR	MCDB	Enth	MCDB	Enth	MCDB	Enth	MCDB	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
16.7	12.2	18.7	15.6	11.4	18.0	14.7	10.7	17.5	50.5	20.1	47.6	19.1	45.4	18.4	21.4

Extreme Annual Design Conditions

Extreme Annual WS			Extreme Annual Temperature				n-Year Return Period Values of Extreme Temperature									
			Mean		Standard Deviation		n=5 years		n=10 years		n=20 years		n=50 years			
1%	2.5%	5%	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	DB	WB
(a)	(b)	(c)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)		
(4) 18.0	15.9	14.0	-27.9	25.9	2.9	2.4	-30.0	27.7	-31.7	29.1	-33.3	30.5	-35.5	32.3		
(5)			N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A		

Monthly Climatic Design Conditions

		Annual	Jan	Feb	Mar	Apr	May	Jun	Jul	Aug	Sep	Oct	Nov	Dec
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
Temperatures, Degree-Days and Degree-Hours	DBAvg	3.8	-9.5	-8.5	-3.4	2.5	7.8	11.3	14.6	16.2	12.9	6.4	0.1	-6.0
	DBStd	9.90	6.67	6.78	5.14	3.47	3.01	2.91	2.97	2.42	3.37	3.62	4.52	7.00
	HDD10.0	2788	606	519	414	226	81	17	2	0	12	121	296	494
	HDD18.3	5331	864	752	673	476	327	212	121	74	164	371	546	753
	CDD10.0	516	0	0	0	1	13	56	144	193	100	8	0	0
	CDD18.3	15	0	0	0	0	0	1	5	8	1	0	0	0
	CDH23.3	29	0	0	0	0	1	4	11	11	2	0	0	0
	CDH26.7	2	0	0	0	0	0	0	1	1	1	0	0	0
Wind	WSAvg	6.9	8.1	7.7	6.7	6.3	5.9	5.8	5.6	5.2	6.8	7.6	8.4	8.4
Precipitation	PrecAvg	879	47	34	42	67	88	100	96	89	104	91	68	53
	PrecMax	1050	112	84	83	196	207	204	170	137	215	167	133	95
	PrecMin	646	11	11	11	18	48	46	57	42	43	34	25	26
	PrecStd	101	19	17	16	33	31	35	29	26	37	34	26	19
Monthly Design Dry Bulb and Mean Coincident Wet Bulb Temperatures	0.4%	DB	2.9	4.4	7.7	14.3	19.2	23.5	25.0	24.6	21.5	16.2	11.3	5.7
		MCWB	0.5	1.6	3.9	8.0	10.9	13.9	17.1	15.8	15.1	11.5	7.3	2.7
	2%	DB	1.7	2.8	5.2	10.6	15.9	20.0	21.9	22.3	19.4	13.6	8.0	4.4
		MCWB	-0.1	0.7	2.3	5.1	9.2	13.7	15.5	15.4	14.5	9.8	5.2	1.9
	5%	DB	0.9	1.9	3.4	8.2	13.8	17.9	20.2	20.6	18.1	12.2	6.7	3.4
		MCWB	-0.7	0.2	1.0	3.7	8.7	12.5	15.3	15.3	14.3	9.1	4.4	1.5
	10%	DB	-0.2	0.6	2.2	6.4	12.1	16.1	18.8	19.4	16.9	10.9	5.6	2.5
		MCWB	-1.7	-1.1	0.2	2.4	7.8	11.8	14.6	15.3	13.9	8.0	3.4	0.8
Monthly Design Wet Bulb and Mean Coincident Dry Bulb Temperatures	0.4%	WB	1.2	2.5	5.6	8.6	13.5	16.4	18.6	19.2	17.3	12.5	8.0	3.5
		MCDB	2.5	3.7	6.8	13.0	17.0	20.2	20.9	21.2	19.5	14.5	10.5	5.0
	2%	WB	0.3	1.2	2.9	5.9	11.1	14.8	17.4	17.8	16.3	10.7	6.1	2.5
		MCDB	1.4	2.4	3.9	9.7	14.2	18.1	20.3	19.7	18.1	12.9	7.9	3.7
	5%	WB	-0.6	0.3	1.3	4.0	9.8	13.7	16.4	17.0	15.3	9.4	4.8	1.8
		MCDB	0.8	1.7	3.0	7.6	12.7	16.8	19.3	19.0	17.3	11.4	6.4	3.2
	10%	WB	-1.8	-1.0	0.4	2.8	8.4	12.5	15.4	16.2	14.2	8.2	3.7	0.8
		MCDB	-0.1	0.5	2.0	5.7	11.4	15.7	18.2	18.4	16.3	10.2	5.6	2.5
Mean Daily Temperature Range	5% DB	MDBR	7.0	7.4	5.7	5.4	6.9	7.9	6.5	4.8	4.4	4.4	4.6	6.2
		MCDBR	6.6	6.2	5.6	8.6	10.3	11.1	8.3	6.5	5.5	6.2	5.0	4.3
		MCWBR	6.5	5.4	4.1	5.2	6.3	6.6	4.7	3.8	4.0	5.0	4.6	4.2
	5% WB	MCDBR	6.3	5.1	5.0	8.0	9.5	9.9	7.6	5.7	4.7	6.0	4.8	4.4
MCWBR		6.5	4.6	4.0	5.3	6.9	7.1	5.3	4.5	4.2	5.3	4.8	4.4	
Clear-Sky Solar Irradiance	taub	0.269	0.271	0.302	0.340	0.361	0.379	0.398	0.374	0.357	0.316	0.290	0.266	
	taud	2.439	2.424	2.402	2.381	2.416	2.397	2.397	2.397	2.462	2.488	2.564	2.538	2.490
	Ebn at Noon	819	902	920	911	900	883	860	867	850	839	791	781	
	Edh at Noon	63	80	97	111	111	114	113	101	90	71	58	54	
All-Sky Solar Radiation	RadAvg	1.06	2.08	3.39	4.72	5.48	5.90	5.95	5.12	3.67	2.17	1.14	0.86	
	RadStd	0.13	0.22	0.30	0.49	0.36	0.41	0.32	0.38	0.25	0.27	0.12	0.09	

Historical Trends

		Heating			Cooling			Degree-Days			
		99% DB		99% DP	1% DB	1% WB	1% DP	HDD10.0	HDD18.3	CDD10.0	CDD18.3
(46) Station Only	DBAvg	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)
(47) Regional (16 neighbors)		N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A

Nomenclature: See separate page